



HITEN REDDY KAMBAM

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Professional Summary

Overall, 11+ years of diverse experience in the field of Data Space, which includes experience in requirement gathering, design and development, and Implementation of various applications as well as experience as a Data Engineer with cloud platforms like **AWS, AZURE, SNOWFLAKE** and **DATABRICKS**. Proficient in big data technologies using Spark for data cleansing, data analysis, structured and unstructured data transformations, data modeling, data warehousing, and data visualizations using PySpark, Spark SQL, Python, SQL, PLSQL, AWS, Snowflake, Data Bricks, IICS, Informatica PowerCenter, Crystal Reports and SAP BO. In addition, I hold a **AWS Solution Architect** certification.

- Having experience in designing and developing **AWS Cloud** solutions, specializing in data migrations, Business Intelligence, **ETL, ELT**, Data Integration, and **BI Reports** development.
- Data Architecture Design: Design and implement the data architecture, including data models, databases, and data pipelines, ensuring efficient data storage, retrieval, and processing.
- Data Integration: Integrated data from various sources, such as transactional systems and external **APIs** into target Database, ensuring data consistency and integrity.
- Data Transformation and ETL: Developed and maintained Extract, Transform, Load (**ETL**) processes to cleanse, transform, and load data into the appropriate formats and structures required by the client.
- **Data Security and Privacy**: Implemented data security measures, such as **encryption** and access controls, to ensure the confidentiality and integrity of sensitive customer data.
- Performance Optimization: Optimized data retrieval and processing performance, implementing indexing strategies, query optimizations, and caching mechanisms to enhance the platform's overall performance.
- Data Quality Assurance: performed data quality checks, validation, and monitoring to ensure the accuracy, completeness, and consistency of data used within the digital bank and digital credit card platform.
- Design and Develop ETL Processes in **AWS Glue** to migrate data from external sources like **S3**, and Text Files into **AWS Redshift**.
- Expertise in AWS services including **S3, EC2, SNS, SQS, RDS, EMR, Kinesis, Lambda, Step Functions, Glue, Redshift, DynamoDB, Elasticsearch, Service Catalog, CloudWatch, and IAM**. Experienced in utilizing AWS utilities such as **EMR, S3, and CloudWatch** to run and monitor Hadoop and Spark jobs on Amazon Web Services (AWS).
- Proficient in working with **Amazon EC2** to provide a solution for computing, query handling, and storage across a wide range of applications.
- Experienced in utilizing **AWS S3** to support data transfer over **SSL** and ensure automatic encryption of data whenever it is uploaded.
- Proficient in using **AWS Glue** to extract, transform, and load data from various sources to target destinations.
- Experience in dealing with **Apache Spark** and Apache Hadoop components like RDDs, Data Frames, HDFS, MapReduce.
- Proficient in working with **Databricks, Spark** and languages such as Python.
- Proficient in responsible for designing, developing, and maintaining data transformation pipelines using DBT, enabling data-driven decision-making for the organization.
- Experience in Developing Spark applications using **Spark - SQL/PYSPARK** in **Databricks** for ETL from multiple files (Parquet, CSV file) formats for analyzing & transforming the data to customer usage patterns.

- Well-versed in **PySpark**, Databricks Clusters, Delta Lake, Unity catalog, Workflows, SQL DWH Endpoint, and Azure Data Factory's Dynamic Expression Language, with a strong understanding of how to use them in data migration, processing, and analysis tasks.
- Proficient in working with **GitHub**, and **GitHub Actions**, and successfully deployed the **AWS DevOps** Release Pipeline with **CDK CloudFormation** Templates and Databricks Notebooks.
- Hands-on experience with Apache Airflow, an open-source platform for orchestrating and scheduling data workflows. Developed and maintained **Airflow DAGs** (Directed Acyclic Graphs) for automating data pipelines, including task dependencies, scheduling, and monitoring.
- Experience in configuring **Spark Streaming** to receive real-time data from **Apache Kafka** and store the stream data to HDFS. and expertise in using **spark-SQL** with various data sources like JSON and Parquet.
- Experienced in Optimizing the Virtual warehouses and the SQL Query in terms of the cost in Snowflake.
- Experienced in Bulk loading from the external stage (**AWS S3**), internal stage to Snowflake cloud using the COPY command.
- Experienced in Loading data into Snowflake tables from the external stage using snow pipe.
- Used import and Export from the internal stage (snowflake) from the External stage (**AWS S3**).
- Writing complex **Snow SQL** scripts in Snowflake cloud data warehouse to business analysis and reporting.
- Experienced in fact dimensional modeling (Star schema, Snowflake schema), transactional modeling, and SCD (Slowly changing dimension).
- Strong **RDBMS** concepts created and maintained Views, Stored Procedures, User-Defined Functions, and System Functions using SQL Server, Oracle and DB2 worked on the design of star and snowflake schemas.

Technical Summary

Programming	Python, PySpark, Scala, Java, Shell Script, Perl, TCL, Power Shell, SQL, JavaScript, HTML, CSS.
Databases	Oracle, MySQL, SQL Server, Mongo DB, Cassandra, Dynamo DB, PostgreSQL, Informatica, Teradata, Cosmos, Neo4j.
Cloud Technologies	AWS (SNS, SQS, Step Function, Event Bridge, Lambda, EC2, EMR, Amazon S3, Kinesis, Firehose, Sagemaker, Athena, Redshift, Glue, DynamoDB, Elasticsearch, CloudWatch, KMS, Cloud Formation, API Gateway, VPC, and IAM), Azure Data Bricks.
Big Data Technologies	Hadoop, MapReduce, HDFS, Sqoop, PIG, Hive, HBase, Oozie, Flume, Nifi, Kafka, Zookeeper, Yarn, Apache Spark, SparkLib
Frameworks	Flask, Django
Tools	Glue Studio, Data Bricks Notebook, PyCharm, Eclipse, DBT, Visual Studio, SQL*Plus, SQL Developer, TOAD, SQL Navigator, Query Analyzer, SQL Server Management Studio, Informatica, Postman, CDK, Schema Change.
Versioning Tools	SVN, Git, GitHub (Version Control)
Network Security	AWS VPC
Database Modelling	Dimension Modelling, ER Modelling, Star Schema Modelling, Snowflake Modelling
Monitoring Tool	Apache Airflow, Agile, Jira, Rally
Visualization/Reporting	Tableau, ggplot2, Matplotlib, SSRS, and Power BI
Web Technologies	HTML, CSS, Bootstrap, JavaScript
Machine Learning	Pandas, SparkML

Professional Experience

Senior Data Engineer
U.S. Bank, Irving, TX

Jan 2023 - Present

Key Responsibilities:

- Collaborating with leads to understanding project requirements and creating functional specifications for **AWS-based data platforms**.
- Led the initiation of a **Data Governance** project to address data security and compliance requirements.
- Implemented data classification policies to categorize data based on sensitivity and regulatory requirements.
- Ensured consistent tagging of data assets to enforce security and compliance standards.
- Established **data lineage** tracking to visualize the flow of data within the organization.
- Conducted impact analysis to assess the consequences of data source changes on downstream processes.
- Implemented data quality improvement measures, including data profiling, data cleansing, and data monitoring.
- Configured **AWS S3** buckets with policies for automatic archiving of infrequently accessed data to storage classes.
- Developed the robust **ETL** pipeline in **AWS Glue** to integrate data from on-premises to cloud and applied transformations using **PySpark** to load the enriched data to Snowflake.
- Developed the Scalable ETL pipeline using PySpark on **AWS Glue**, loaded the enriched data into **Snowflake**.
- Used CI/CD pipelines to Implement AWS and snowflake infra using **GitHub actions**.
- Implemented **AWS** and **Snowflake** resources using python code.
- Used **CDK stack** to implement the AWS infrastructure and Schema change to implement the snowflake infrastructure.
- Implemented **Sage maker** backup process in SBD analytics account and Life Exploratory account as part of remediation work.
- Implemented **Sage maker** restore process in Life Exploratory account Dev and documented the restoration process.
- According to enterprise standards certified **GitHub repos**. Implemented integration test cases in snowflake and unit test cases for **AWS constructs**.
- Data product in-force lives: Developed new load patterns in snowflake to fetch Policy, Benefits, Members data from on Prem source systems (Developed lightly dependent pipelines whose sources can be switched in future)
- Data Product Eligibility Lives and Participation rate: created pipelines to fetch eligible lives data from capture tables and developed a logic to correction for claims status.
- Claims report correction: Existing logic correction for claims status.
- Business Objects: **SAP BO** Server upgrades are done at the server side; BO components and reports testing is done to verify/ensure all the functionalities are working fine after the upgrade.
- Worked on adding intraday activities and removing duplicates in **CLAIMS_TASK_WORKFLOW** curated table. Created **CLAIMS_STATUS_TRACKING_CURATED** table for compliance ask.
- Contributed to strengthening data security by assigning the transition from an outdated, hardcoded snowflake process account. I focused on updating the source code to remove reference of old account, ensuring a smooth shift to enhanced, secure credential management. These efforts help to reduce security risk and aligned with our data protection goals.
- Supported our compliance and quality efforts by preparing for an upcoming audit on **GitHub** repositories. I helped to ensure our repos align with enterprise standards by gathering necessary documentation and proving proof of adherence. This contribution strengthens our commitment to maintaining secure and standardized development practices.
- Developed **AWS Lambda** python functions using S3 triggers to automate workflows.
- Developed data pipelines using Pyspark/Python in **AWS Glue** Interactive Notebook, leveraging Spark DataFrames for data extraction, transformation, and persistence within the AWS ecosystem.
- Provided expert guidance on AWS best practices, particularly regarding security and **EMR** configurations.
- Experienced in handling large datasets using Partitions, Spark in Memory capabilities, Broadcasts in Spark, Effective & efficient Joins, Transformations and other during ingestion process itself.
- Exploring with Spark, improving performance and optimization of the existing algorithms in Hadoop using Spark Context, **Spark-SQL**, Data Frame, and Pair RDD's.
- Implemented Spark applications using Spark - SQL/spark/data frame in **AWS Glue** for data extraction,

transformation, and aggregation from multiple file formats (csv, Jason, parquet, RDBMS) for analyzing & transforming the data to uncover insights into the customer.

Environment: AWS EMR, S3, Lambda, Step Functions, CFT's, SNS, SQS, Snowflake, Kinesis, CloudWatch, EMRFS, LINUX, Hive, PySpark, Databricks, Python, Snowflake, Tableau.

Senior Data Engineer

MERCK PHARMA, Branchburg, NJ

Jan 2021 – Dec 2022

Key Responsibilities:

- Developed Data pipelines using Python for medical image pre-processing, Training, and Testing.
- Developed **PySpark** Applications by using python and Implemented Apache PySpark data processing project to handle data from various RDBMS and Streaming sources.
- Develop a data set process for data mining and data modeling and recommend the ways to improve data quality, efficiency and reliability.
- Built data pipelines for reporting, alerting, and data mining. Experienced with table design and data management using **HDFS, Hive, Impala, Sqoop, MySQL, and Kafka**.
- Worked on **Apache Nifi** to automate the data movement between RDBMS and HDFS.
- Created **shell scripts** to handle various jobs like Map Reduce, Hive, Pig, Spark etc., based on the requirement.
- Experience on **Spark-SQL** for processing the large amount of structured data.
- Extract Transform and Load data from Sources Systems to AWS Data storage services using a combination of **AWS S3, EC2, SNS, SQS, RDS, EMR, Kinesis, Lambda, Step Functions, Glue, Redshift, DynamoDB, Elasticsearch, Service Catalog, CloudWatch, and IAM**.
- Handled importing of data from various data sources, performed data control checks using PySpark and loaded data into **HDFS**.
- Involved in converting **Hive/SQL** queries into PySpark transformations using Spark RDD, python.
- Used PySpark SQL to Load JSON data and create Schema RDD and loaded it into **Hive** Tables and handled structured data using Spark SQL.
- Developed **PySpark** Programs using python and performed transformations and actions on RDD's.
- Imported data from **AWS S3** into Spark RDD, Performed transformations and actions on RDD's.
- Developed JSON scripts for deploying the pipeline **AWS data factory** that processes the data using the SQL activity.
- Used PySpark and Spark SQL to read the parquet data and create the tables in hive using the **python API**.
- Implemented **PySpark** using python and utilizing Data frames and PySpark SQL API for faster processing of data.
- Developed python scripts, UDFs using both Data frames/SQL/Data sets and RDD/Map Reduce in **Spark 1.6** for data aggregation, queries and writing data back into OLTP system through Sqoop.
- Experienced in handling large datasets using Partitions, **PySpark** in Memory capabilities, Broadcasts in PySpark, effective & efficient Joins, Transformations and other during ingestion process itself.
- Processing the schema oriented and non-schema-oriented data using python and **Spark**.
- Involved in writing live Real-time Processing and core jobs using **Spark Streaming** with Kafka as a data pipeline system.
- Used **Spark** API over Cloudera Hadoop YARN to perform analytics on data in HDFS.
- Worked on streaming pipeline that uses **PySpark** to read data from Kafka, transform it and write it to **HDFS**.
- Designed Data Marts by following **Star Schema** and **Snowflake Schema** Methodology, using industry leading Data Modeling tools.
- Worked on Snowflake database on queries and writing Stored Procedures for normalization.
- Worked with **Snowflake's** stored procedures, used procedures with corresponding DDL statements, used JavaScript API to easily wrap and execute numerous SQL queries.

Environment: Snowflake, Python, AWS, DMS, JIRA, GIT, GITHUB Actions, Spark, Pyspark, Snowpark, Confluence, Informatica PowerCenter, Oracle, Snow SQL, Snow Pipe, Agile, OLTP, OLAP, JIRA.

Key Responsibilities:

- Worked on building a centralized **Data Lake** on **AWS Cloud** utilizing primary services like **S3, EMR, Redshift, and Athena**.
- Lead the team in deploying methodologies, ensuring cohesiveness and adherence to best practices.
- Lead data curation efforts to ensure high-quality datasets, including **data validation, cleansing, and monitoring processes**.
- Involved in designing, developing, and maintaining **SAP BW-related flows** and reports, which are used for **business intelligence** and **data analytics** purposes. These flows and reports may include data extraction, transformation, and loading (ETL) processes, data modeling, and report development using **SAP BW tools**.
- Experience in creating **HANA** views, which are virtual data models that can be used to simplify data access and enable faster query processing. **HANA** views can be used for reporting, analytics, and other data-driven applications.
- Worked on migrating datasets and **ETL workloads** with Scala from On-prem to **AWS Cloud services**.
- Extensive experience in utilizing the **ETL Process** for designing and building very large-scale data using **Apache Spark**.
- Migrating the data from the **local Teradata data warehouse** to **AWS S3 data lakes**.
- Built a series of **Spark Applications** and **Hive scripts** to produce various analytical datasets needed for digital marketing teams.
- Worked extensively on building and automating **data pipelines** and moving **terabytes** of data from existing data warehouses to the cloud.
- Responsible for **Data Ingestion** projects to inject the data into a data lake using multiple data source systems using **Talend Bigdata**.
- Worked extensively on fine-tuning **spark applications** and providing production support to various pipelines running in production.
- Developed Python code to gather the data from **HBase** and designed the solution to implement using **PySpark**.
- Developed and optimized Python-based **ETL pipelines** in both legacy and distributed environments.
- Developed Spark with Python-based pipelines using Spark data frame operations to load data to **EDL** using **EMR** for jobs execution & **AWS S3** as a storage layer.
- Worked closely with business teams and data science teams and ensured all the requirements were translated accurately into our data pipelines.
- Migration of on-premises data (SQL Server / MongoDB) to **Azure Data Lake Store (ADLS)** using **Azure Data Factory (ADF V1/V2)**.
- Worked on the full spectrum of **Data Engineering pipelines, data ingestion, data transformations, and data Analysis/consumption** with Python.
- Extracted the data from **AWS Aurora Databases** for big data processing.
- Developed **AWS lambdas** using Python & **Step functions** to **orchestrate data pipelines**.
- Worked on automating the infrastructure setup, launching, and termination of **EMR clusters**, etc.,
- Created **Hive** external tables on top of datasets loaded in **AWS S3 buckets** and created various hive scripts to produce a series of aggregated datasets for downstream analysis.
- Used Scala data pipelines to perform transformations on the **EMR clusters** and loaded the transformed data into S3 and from S3 into **Redshift**.
- Worked on creating Kafka producers using **Kafka Java Producer API** for connecting to external Rest live stream applications and producing messages to **Kafka** topic.

Environment: AWS S3, Lambda, SNS, EMR, Kinesis, Sage maker, DynamoDB, Elasticsearch, CloudWatch, Redshift, Aurora, Athena, Glue, Talend, Spark, Python, Java, Hive, Kafka.

- Configured **AWS S3 buckets** with policies for automatic archiving of infrequently accessed data to storage classes.
- Designed and implemented real-time data streaming solutions using **Solace** & utilized **Solace messaging** capabilities to enable timely data delivery and event-driven processing.

- Configured **Solace** message queues and topics for efficient and reliable data streaming & collaborated with the development team to ensure seamless integration of **Solace within the project architecture**.
- Demonstrated a commitment to continuous improvement by exploring and adopting new features and capabilities of **Solace** and **Cloudera**.
- Implemented **DQ/data validation** check using **Spark** on incoming messages/beacons and then stored the results in **Elastic search**. These DQ results were then shown on **Kibana** through Elastic search integration.
- Implemented **CI/CD** by automating the spark jobs build and deployment using **Jenkins** and **Ansible**.
- Developed the **Python** scripts using **Boto3** to supplement automation provided by **Terraform** for tasks such as encrypting **EBS volumes backing AMLs** and scheduling **Lambda** functions for routine **AWS** tasks.
- Developed scalable data integration pipelines to transfer data from **AWS S3 bucket** to **AWS Redshift** database using **Python** and **AWS Glue**.
- Built **Apache airflow DAGs** to export the data to **AWS S3 buckets** by triggering to invoke an **AWS lambda** function.
- Analyzing files from the S3 data lake using **AWS Athena** and **AWS Glue** without importing the data into a database.
- Created ad-hoc tables to add schema and structure data into **AWS S3 bucket** using lambda function and performed data validation, filtering, sorting, and transformations for every data change in a **Dynamo DB** table and load the transformed data to **Postgres database**.
- Implemented Hadoop file system (**HDFS**), AWS S3 storage, and Bigdata formats including **Parquet** and **AVRO JSON** for enterprise data lake.
- Worked on **ETL Migration** services by developing and deploying AWS Lambda functions for generating a serverless data pipeline which can be loaded to **Glue Catalog** and can be queried from **AWS Athena**.
- Optimized **Druid query performance** by tuning memory settings, **caching strategies**, and **segment size**.
- Designed and implemented a real-time analytics system using **Druid**, **Apache Kafka**, and **Apache Flink** to analyze user behavior on our e-commerce platform.
- Implemented **AWS Elasticsearch** for storing massive datasets in a single cluster for extensive log analysis.
- Automated ETL operations using **Apache Airflow**, optimized queries, and fine-tuned performance in **AWS Redshift** for large dataset migration.
- Configured **AWS Redshift clusters**, spectrums for querying, and data sharing for data transfer between clusters.
- The collected data in **InfluxDB** was viewed in **Grafana** – where the metrics can be queried and visualized easily.
- Worked on analyzing **Hadoop cluster** and different big data analytic tools including **Hive**, **HBase**.
- Highly proficient in developing **lambda functions** to automate tasks on AWS using **CloudWatch** triggers, **S3** events as well as **kinesis streams**.
- Knowledge of **Airflow's** built-in features for managing and scaling data pipelines, such as pooling and task prioritization.
- Proficient in creating calculated fields, parameters, and groups to generate custom views and analysis, sharing reports and dashboards with colleagues and clients, including publishing to **Tableau Server** or **Tableau Online**.

Environment: Apache Spark, Python, SQL, AWS S3, Athena, SNS, SQS, Lambda, CloudWatch, Dynamo DB, Redshift, AWS Glue, EMR, Apache Kafka, Apache Hadoop, HDFS, HBase, Map Reduce, Hive, Pig, Sqoop, Flume, Shell Scripting, SFTP, Tableau.

Junior Data Engineer
Discover Bank, Hyderabad, India

June 2013 – Sep 2016

Key Responsibilities:

- Implemented report generation onto PowerBI using Synapse databases.
- Extracted data from Oracle Golden Gate tables, flat files using Informatica ETL mappings and loaded to Data Mart.
- Designed and developed ETL jobs for the data management teams in various projects.
- Build and maintain SQL scripts, Indexes, and complex queries for data analysis and extraction.
- Developed complex Source qualifier queries, Lookup override queries to improve the performance of the ETL load jobs.
- Creation of reports using prompts/ combined queries, report level filters.
- Used Sagemaker as dev endpoint for the glue development.
- Authored Spark Jobs for data filtering and data transforming through Pyspark data frames in both AWS glue and databricks.
- Imported data using Sqoop to load data from MySQL to HDFS on regular basis.
- Defined and modified standard design patterned ETL frameworks, Data Model standards guidelines and ETL best practices.
- Performed detailed data investigation and analysis of known data quality issues in related databases through SQL.

Environment: Spark, Hadoop, Hive, HBase, Oracle, Spark-SQL, PowerBI, Pyspark, Python.

Achievements:

- Got appreciation from top leaders for successfully implementing migration project from on Prem to Snowflake using AWS and Databricks in U.S Bank.
- Received Two Times #GO ABOVE AWARDS at Albertsons Pleasanton.
- “Star of The Month” appreciation at MERCK PHARMA.
- “On Spot” Award at MERCK PHARMA.

Education:

Bachelor’s Degree in Computer Science and Engineering
Jawaharlal Nehru Technological University, Anantapur, India.

Aug 2009 – April 2013